

NETWORKING NOTES

Module 2 • Category-Wise Practical Guide

Linux, Networking & Cloud Infrastructure

Practical notes covering IP addressing, DNS, ports, firewalls and Azure Network Security Groups (NSGs).

Goal: Understand how traffic moves from a client to a Linux VM and how network controls affect that traffic.

Md Shahil

1. Networking Fundamentals

Think of a connection as: name resolution → destination IP → protocol/port → network security controls → host → application.

Client | | DNS v Destination IP | | TCP/UDP + port v NSG / Firewall | v Linux VM / Application

A working connection normally requires a reachable destination, the correct protocol/port, and rules that permit the traffic.

2. IP Addressing

An IP address identifies a network interface or endpoint so traffic can reach the intended destination. IPv4 uses four decimal octets, for example **10.0.1.10**.

Type	Purpose	Example
Private IP	Used inside private network scopes such as a VNet/subnet.	10.0.1.10
Public IP	Internet-routable endpoint when configured for a resource.	203.0.113.10

CIDR notation describes a network and its prefix length.

Example	Meaning
10.0.0.0/16	Larger private network/address space.
10.0.1.0/24	Smaller subnet inside that network.
10.0.1.10	Individual private IP in the subnet.

Why subnet? Subnets logically separate resources and allow different routing and security rules.

3. DNS — Domain Name System

DNS translates human-readable names into IP addresses. It lets users access services by name instead of remembering an IP.

www.example.com → DNS → 203.0.113.10

Record	Common purpose	Example
A	Hostname to IPv4 address.	app.example.com → 203.0.113.10
AAAA	Hostname to IPv6 address.	app.example.com → IPv6
CNAME	One hostname to another hostname.	www → app.example.com
MX	Mail-server information.	example.com → mail server
TXT	Text for verification and other uses.	domain verification

Key point: DNS resolves a name; it does not itself decide whether a network connection is allowed.

4. Ports & Protocols

A port identifies a service endpoint on a host. TCP and UDP are common transport protocols.

Port	Service	Protocol	Typical use
22	SSH	TCP	Remote Linux administration
53	DNS	UDP/TCP	DNS operations
80	HTTP	TCP	Web traffic
443	HTTPS	TCP	Encrypted web traffic
3389	RDP	TCP	Windows remote desktop

A service listening on a port does not automatically make it reachable from the internet; routing and security controls also matter.

```
sudo ss -lntp sudo ss -tulpn
```

5. Firewalls

A firewall filters network traffic according to rules. Rules may evaluate source, destination, protocol and port.

Layer	Example	Purpose
Cloud/network	Azure NSG	Controls allowed/denied network traffic.
Host	UFW / nftables	Controls traffic at the Linux OS level.
Application	Nginx configuration	Controls application behavior after traffic arrives.

6. Azure Network Security Group (NSG)

An Azure NSG contains inbound and outbound security rules. Rules can use source, destination, protocol, port and priority. An NSG can be associated with a subnet or network interface.

Example: SSH access to a Linux VM

Field	Example	Meaning
Direction	Inbound	Traffic entering the protected resource/network interface.
Source	Trusted IP / required source	Where the connection originates.
Destination	VM / network interface	Target resource.
Protocol	TCP	Transport protocol.
Destination port	22	SSH service port.
Action	Allow	Permit matching traffic.
Priority	Suitable unique priority	Lower numerical priority is evaluated before higher numerical priority.

Security practice: Restrict SSH to trusted sources where practical instead of allowing it from any source.

Example: Nginx HTTP access

Direction	Inbound
Protocol	TCP
Destination port	80
Purpose	Allow HTTP requests when HTTP is required.

HTTPS normally uses port 443.

7. Azure Linux VM Traffic Flow

Internet / Client | | DNS v Public IP / Endpoint | | TCP 22 or 80/443 v Azure Network Controls | v VM
NIC / Subnet | v Linux VM | +-- SSH :22 +-- Nginx :80 / :443

The exact path depends on the architecture. An Application Gateway or load balancer may sit in front of the VM instead of exposing the VM directly.

8. Networking Troubleshooting

Step	Question	Useful check
1	Does hostname resolve?	DNS lookup / browser resolution
2	Is destination IP correct?	Check DNS result and Azure IP configuration
3	Is required port used?	22 SSH; 80/443 web

Step	Question	Useful check
4	Does Azure allow it?	Check NSG direction, source, destination, port and priority
5	Does Linux allow it?	Check UFW/nftables if configured
6	Is service running?	systemctl status nginx
7	Is service listening?	sudo ss -lntp
8	Does app respond locally?	curl http://localhost

Useful Linux networking commands:

`ip addr ip route ping <destination> curl http://localhost sudo ss -lntp`

A failed ping does not automatically prove a web service is unreachable because ICMP may be blocked while TCP 80/443 is allowed.

9. Module 2 Practical Evidence

Evidence	What it demonstrates
Azure VM screenshot	Linux VM deployed in Azure.
VNet/subnet/NSG screenshot	Azure network structure and traffic-control configuration.
SSH success screenshot	Successful TCP/22 access and Linux login.
Nginx status screenshot	Web service installed/running.
Browser/curl screenshot	Connectivity/application response.
Networking diagram	Understanding of the client-to-workload traffic path.

10. Quick Interview Revision

What is an IP address?

An address used to identify a network endpoint/interface for communication.

Private vs public IP?

Private IPs are used within private network scopes; public IPs provide internet-routable endpoints when configured.

What does DNS do?

It resolves names such as app.example.com to IP addresses.

What is a port?

A numbered service endpoint used with network protocols such as TCP or UDP.

Why port 22?

TCP port 22 is conventionally used by SSH.

Why port 80?

TCP port 80 is conventionally used by HTTP.

What does a firewall do?

It filters traffic according to configured rules.

What is an Azure NSG?

An Azure network security control containing rules that allow or deny network traffic.

Why can SSH fail even when the VM is running?

The IP, route, NSG, host firewall, SSH service, or credentials may be the issue.

How do you check Nginx?

Use systemctl status nginx, ss to check listening sockets, and curl to test locally.

11. Suggested GitHub Structure

Module-2-Linux-Networking-Azure/ ■■■ README.md ■■■ linux/ ■ ■■■ linux-notes.md ■■■ networking/ ■ ■■■ networking-notes.md ■■■ diagrams/ ■ ■■■ linux-filesystem.png ■ ■■■ networking-traffic-flow.png ■■■ screenshots/ ■■■ azure-linux-vm.png ■■■ nsg-config.png ■■■ ssh-success.png ■■■ nginx-status.png ■■■ nginx-running.png

Module 2 coverage: IP addressing, DNS, ports, firewalls and Azure NSG. Use your own lab's actual IPs, Azure resources and screenshots in the repository.